

Financial and Time Series Econometrics

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1 Part 1: Data Description and Stylized Facts

In this section, we study the *SPDR Gold Shares ETF* (GLD), which is an exchange traded fund that tracks the price of gold and is actively traded in financial markets. The data are downloaded from Yahoo Finance and consist of adjusted closing prices, meaning that they already take into account dividends and stock splits. The sample runs from January 2015 to December 2025. In total, there are 2,761 daily observations and 574 weekly observations. This long sample period makes it possible to analyze the main statistical features of the series across different market phases.

We work with both daily and weekly data. Daily prices are taken from the adjusted closing price series. Weekly prices are obtained by converting the daily data to weekly frequency and retaining the weekly adjusted closing value. Returns are computed as log differences of prices, which is standard practice in financial time series analysis.

1.1 Graphical analysis

Figures 1 and 2 show the evolution of GLD prices and returns at both daily and weekly frequencies. Looking at the price series, there is a clear upward trend over time. Prices increase from values around 100 at the beginning of the sample to above 400 towards the end. This visual pattern suggests that prices may be non-stationary in levels.

Returns behave very differently from prices. They fluctuate around zero and do not exhibit any visible trend over time. For daily returns, there are periods with large movements followed by calmer periods, which is a typical example of volatility clustering. Weekly returns appear smoother compared to daily ones, but they still show episodes of higher volatility. The histograms of returns indicate that the distributions are more peaked and have heavier tails than the normal distribution, especially for daily returns.

1.2 Summary statistics

Table 1: Summary Statistics of GLD Prices

Frequency	N	Start	End	Min	Max	Mean	SD
Daily	2761	2015-01-02	2025-12-23	100.50	413.64	165.3198	59.2529
Weekly	574	2015-01-02	2025-12-23	101.25	413.64	165.6824	59.8635

Prices are adjusted closing prices. Weekly prices are weekly closes obtained by aggregating adjusted daily prices.

Table 1 reports summary statistics for prices. The average daily adjusted price is around 165, with a standard deviation close to 59. Weekly prices show similar magnitudes. The difference between minimum

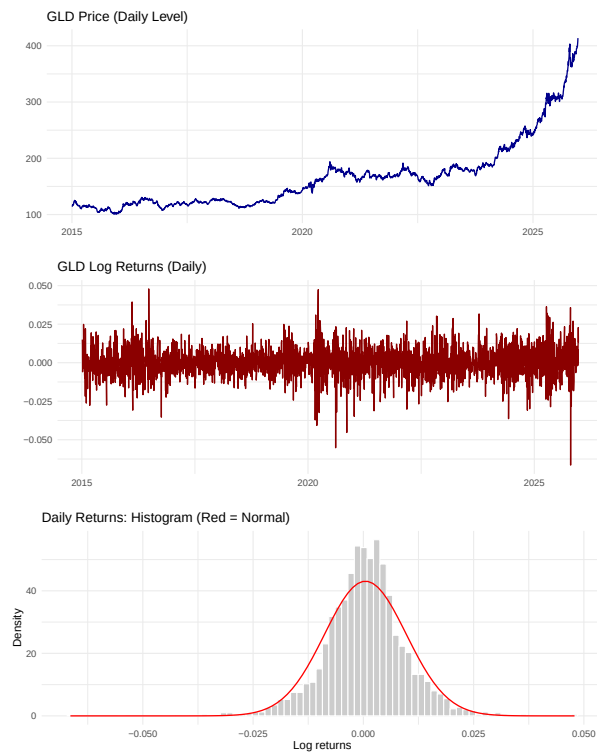


Figure 1: Daily Price and Returns

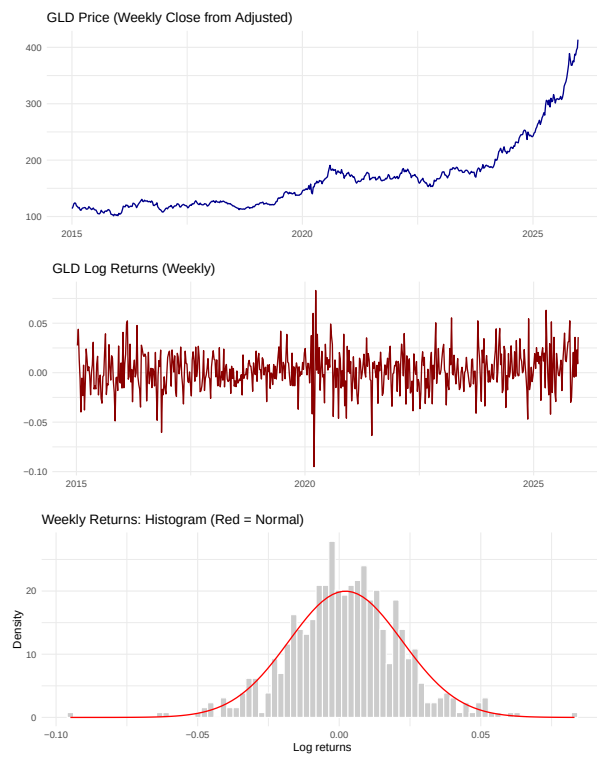


Figure 2: Weekly Price and Returns

and maximum prices is quite large, which reflects the strong long-run variation in gold prices over the sample period.

Table 2: Summary Statistics and Stylized Facts of GLD Returns

Freq.	N	Mean	SD	Skew	Kurt	JB p	LB(r) p	LB(r^2) p	ARCH p
Daily	2760	0.000467	0.009260	-0.218	6.242	< 0.001	0.263	< 0.001	< 0.001
Weekly	573	0.002248	0.019967	-0.069	4.531	< 0.001	0.837	< 0.001	< 0.001

JB is the Jarque–Bera normality test. LB(r) and LB(r^2) are Ljung–Box tests on returns and squared returns. ARCH is the Engle ARCH LM test. Entries are p-values.

Table 2 presents summary statistics for returns. Daily returns have a mean close to zero (0.00046) and a standard deviation of about 0.93%. Weekly returns have a higher mean of 0.22% and also higher volatility, around 1.99%. Skewness is slightly negative for both frequencies. Kurtosis is clearly above 3, equal to 6.25 for daily returns and 4.55 for weekly returns, providing evidence of fat tails, especially at the daily frequency.

1.3 Normality

To formally test for normality, we use the Jarque–Bera test. For daily returns, the test statistic is 1230.4 with a p-value smaller than 0.001, so normality is strongly rejected. Weekly returns also reject normality, with a Jarque–Bera statistic of 56.405 and a p-value below 0.001. These results confirm that GLD returns do not follow a Gaussian distribution.

1.4 Autocorrelation and volatility clustering

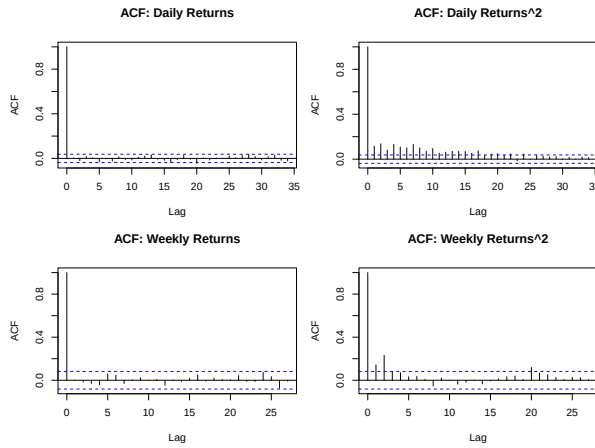


Figure 3: Autocorrelation functions of daily and weekly returns and squared returns

Serial dependence is analyzed using Ljung–Box tests applied to returns and squared returns. Figure 3 shows the sample autocorrelation functions for daily and weekly returns and their squares. For daily returns, the Ljung–Box statistic at lag 20 is 23.54 with a p-value of 0.263, so the null hypothesis of no autocorrelation cannot be rejected. A similar result holds for weekly returns, where the statistic at lag 10 is 5.73 with a p-value of 0.837. Consistent with the ACF plots in Figure 3, raw returns display little serial correlation.

However, the results change significantly when considering squared returns. For daily squared returns, the Ljung–Box statistic is 424.9 with a p-value smaller than 0.001. For weekly squared returns,

the statistic is 52.74 with a p-value also below 0.001. This is clearly visible in Figure 3, where several autocorrelations of squared returns lie outside the confidence bands, providing strong evidence of volatility clustering.

These findings are further supported by ARCH LM tests. For daily returns, the ARCH test statistic is 171 using 12 lags, with a p-value below 0.001. For weekly returns, the statistic is 39.59 using 6 lags, again with a p-value smaller than 0.001. Therefore, the null hypothesis of homoskedasticity is clearly rejected at both frequencies.

1.5 Stationarity

Stationarity is tested using the Augmented Dickey–Fuller test. For daily prices, the ADF statistic is 3.11 with a p-value of 0.99, and for weekly prices it is 3.38 with the same p-value. In both cases, the null hypothesis of a unit root cannot be rejected, indicating that price levels are non-stationary.

On the other hand, returns are stationary. The ADF statistic for daily returns is -13.75 with a p-value of 0.01, while for weekly returns it is -7.77 with a p-value of 0.01. Thus, the unit root hypothesis is strongly rejected for returns at both frequencies.

1.6 Discussion

Overall, the GLD series displays the main stylized facts commonly observed in financial time series. Prices are non-stationary, while returns are stationary. Return distributions are fat-tailed and clearly non-normal. Raw returns exhibit little autocorrelation, but squared returns show strong dependence over time. Volatility clustering and conditional heteroskedasticity are evident in both daily and weekly data. These results suggest that models with time-varying volatility, such as GARCH-type models, are appropriate and will be considered in the next part of the assignment.

2 Part 2: GARCH estimation, volatility forecasting and Value at Risk

In this part we model the conditional volatility of GLD returns using GARCH-type models and evaluate their forecasting and risk performance. We work with both daily and weekly log-returns. As required by the assignment, each return series is split into two subsamples: the first two-thirds are used for estimation, while the remaining one-third is reserved for out-of-sample evaluation. This split is implemented separately for daily and weekly data.

2.1 Model selection and estimation

Table 3: Model selection based on AIC

Model	Daily returns	Weekly returns
sGARCH(1,1) – Normal	-6.717	-5.087
sGARCH(1,1) – Student-t	-6.783	-5.113
GJR-GARCH(1,1) – Student-t	-6.789	-5.120

To select an appropriate volatility specification, we estimate several GARCH-type models on the training sample and compare them using the Akaike Information Criterion (AIC). In particular, we consider a standard GARCH(1,1) model with Normal innovations, a GARCH(1,1) model with Student-t innovations, and a GJR-GARCH(1,1) model with Student-t innovations, which allows for asymmetric

responses of volatility to shocks. Table 3 reports the AIC values for all candidate models at both frequencies.

For daily returns, the GJR-GARCH model with Student-t innovations achieves the lowest AIC value (-6.789), outperforming both the standard GARCH with Normal innovations (-6.717) and the standard GARCH with Student-t innovations (-6.783). A similar pattern is observed for weekly returns, where the GJR-GARCH-t specification again yields the lowest AIC (-5.120), compared to -5.087 for the Gaussian GARCH and -5.113 for the Student-t GARCH. Based on these results, we select the GJR-GARCH(1,1) with Student-t innovations as the preferred model at both frequencies.

Table 4: Estimated GJR-GARCH(1,1) parameters

Parameter	Daily	Weekly
μ	0.00035	0.00141
ω	1.33×10^{-6}	5.01×10^{-6}
α	0.075	0.092
β	0.938	0.943
γ	-0.054	-0.091
Shape (Student-t)	5.25	8.80

The estimated coefficients of the selected models are reported in Table 4. The estimates indicate strong volatility persistence in both cases. For daily data, the estimated α and β coefficients are approximately 0.075 and 0.938, implying a sum close to one. For weekly data, the corresponding estimates are around 0.092 and 0.943, again indicating highly persistent volatility dynamics. The asymmetry parameter in the GJR specification is negative for both daily and weekly returns, suggesting that negative shocks have a stronger impact on future volatility than positive shocks of the same magnitude. In addition, the estimated shape parameters of the Student-t distribution are finite, with a value of about 5.25 for daily returns and 8.80 for weekly returns, confirming the presence of fat tails, especially at the daily frequency.

2.2 Specification analysis and residual diagnostics

After estimation, we assess the adequacy of the selected models by analyzing standardized residuals. The results of the residual diagnostics are summarized in Appendix. For daily returns, the Ljung–Box test applied to standardized residuals yields a p-value of about 0.51, while the test applied to squared standardized residuals yields a p-value of about 0.64. The ARCH LM test also does not reject the null hypothesis of no remaining ARCH effects, with a p-value of about 0.65. Overall, these results suggest that the daily GJR-GARCH-t model captures the main features of conditional heteroskedasticity reasonably well.

For weekly returns, the diagnostic results are weaker. While the Ljung–Box test on standardized residuals shows no evidence of serial correlation, the test on squared standardized residuals yields a p-value around 0.02, and the ARCH LM test is borderline with a p-value slightly above 0.05. This indicates that some remaining conditional heteroskedasticity may still be present at the weekly frequency, which may be partly explained by the smaller sample size and lower information content of weekly data.

2.3 Conditional volatility forecasting

Using the models estimated on the training sample, we compute forecasts of conditional volatility. For daily returns, we generate one-step ahead and five-step ahead forecasts. The one-step ahead forecast of daily volatility is approximately 0.0087, and the five-step ahead forecasts show a very gradual increase, reflecting the high persistence of the estimated volatility process. The full set of multi-step daily volatility

forecasts is reported in Appendix. For weekly returns, the one-step ahead volatility forecast is about 0.0182, which is higher than the daily forecast, as expected due to the longer aggregation period.

2.4 Value at Risk and exceedance analysis

We next compute one-step ahead Value at Risk at the 1% and 5% levels using the models estimated on the training sample. Model parameters are fixed at their estimated values, and the return series is filtered to obtain conditional means and volatilities for the test period. Value at Risk is computed as the conditional mean plus the conditional volatility multiplied by the appropriate quantile of the Student-t distribution.

Table 5: Value at Risk backtesting results

Level	Daily		Weekly	
	Exceedances	Kupiec p -value	Exceedances	Kupiec p -value
1%	8 (exp. 9)	0.684	1 (exp. 2)	0.467
5%	45 (exp. 46)	0.879	8 (exp. 10)	0.597

Table 5 reports the exceedance results and the Kupiec unconditional coverage tests. For daily returns, the expected number of exceedances at the 1% level is 9, while the observed number is 8, corresponding to a failure rate of about 0.87%. At the 5% level, the expected number of exceedances is 46, and the observed number is 45, corresponding to a failure rate of about 4.89%. In both cases, the Kupiec test does not reject the null hypothesis of correct coverage.

For weekly returns, the expected number of exceedances is 2 at the 1% level and 10 at the 5% level. The observed numbers are 1 and 8, respectively. The corresponding failure rates remain close to the theoretical levels, and the Kupiec tests again do not reject correct coverage. Although the weekly test sample is relatively small, these results suggest that the Value at Risk forecasts are reasonably well calibrated.

Overall, the empirical results indicate that the GJR-GARCH model with Student-t innovations provides a good description of the volatility dynamics of GLD returns and delivers Value at Risk forecasts with satisfactory out-of-sample performance at both daily and weekly frequencies.

3 Part 3: Bootstrap hypothesis testing for ARCH(1)

3.1 Bootstrap algorithm

To approximate the finite-sample distribution of LR_n , we adopt a recursive bootstrap procedure. This approach is natural for ARCH models, as it preserves the conditional heteroskedastic structure of the data generating process.

We proceed as follows. First, we estimate the restricted model under the null hypothesis $H_0 : \alpha = \bar{\alpha}$. This yields the restricted estimator $\tilde{\omega}_n$ and the fitted conditional variances

$$\tilde{\sigma}_t^2 = \tilde{\omega}_n + \bar{\alpha}x_{t-1}^2.$$

Based on these estimates, we compute the standardized residuals

$$\tilde{z}_t = \frac{x_t}{\tilde{\sigma}_t}.$$

The residuals are centered to enforce zero mean,

$$\tilde{z}_t^c = \tilde{z}_t - \bar{\tilde{z}},$$

where $\bar{\tilde{z}}$ denotes the sample mean of $\tilde{z}_t$. A bootstrap sample of innovations $\{z_t^*\}_{t=1}^n$ is then generated by resampling with replacement from $\{\tilde{z}_t^c\}$.

Using these bootstrap innovations, we generate a bootstrap sample $\{x_t^*\}$ recursively under the null hypothesis. The bootstrap conditional variance is defined as

$$\sigma_t^{*2} = \tilde{\omega}_n + \bar{\alpha} x_{t-1}^{*2},$$

and the bootstrap observations are given by

$$x_t^* = \sigma_t^* z_t^*.$$

The initial value σ_0^{*2} is taken as given, for instance equal to the first observed squared return.

For each bootstrap sample $\{x_t^*\}$, we re-estimate both the unrestricted and restricted models and compute the bootstrap likelihood ratio statistic

$$LR_n^* = 2 \left(l_n^{(1)*} - l_n^{(0)*} \right),$$

where $l_n^{(1)*}$ and $l_n^{(0)*}$ denote the maximized bootstrap log-likelihoods of the unrestricted and restricted models, respectively.

This procedure is repeated B times, where B is sufficiently large. The bootstrap p -value is computed as

$$p^* = \frac{1}{B} \sum_{b=1}^B \mathbf{1} \{ LR_{n,b}^* \geq LR_n \}.$$

3.2 Boundary case: $\bar{\alpha} = \alpha_0 = 0$

When the true parameter α_0 coincides with the null value $\bar{\alpha} = 0$, the null hypothesis lies on the boundary of the parameter space, since the parameter α is constrained to be non-negative. Maximum likelihood theory relies on the assumption that parameters lie in the interior of the parameter space, an assumption that is violated here. As a result, standard likelihood-based asymptotic theory does not apply, and the limiting distribution of the likelihood ratio statistic is nonstandard (typically not the usual χ^2 limit).

As discussed in Andrews [1], bootstrap procedures may fail to provide a valid approximation in such boundary settings. Consequently, likelihood ratio tests based on standard bootstrap critical values can exhibit size distortions, and the resulting p -values need not be uniformly distributed under the null hypothesis.

References

- [1] Andrews, D. W. K. (2000). Inconsistency of the bootstrap when a parameter is on the boundary of the parameter space. *Econometrica*, 68(2), 399–405.

A Additional Statistical Tests

Table 6: Ljung–Box Tests for Serial Correlation

Frequency	Series	Lags	Test statistic	p-value
Daily	Returns	20	23.54	0.263
Daily	Squared returns	20	424.90	< 0.001
Weekly	Returns	10	5.73	0.837
Weekly	Squared returns	10	52.74	< 0.001

Table 7: ARCH LM Tests for Conditional Heteroskedasticity

Frequency	Lags	Test statistic	p-value
Daily	12	171.09	< 0.001
Weekly	6	39.59	< 0.001

Table 8: Jarque–Bera Tests for Normality

Frequency	Test statistic	Degrees of freedom	p-value
Daily	1230.4	2	< 0.001
Weekly	56.41	2	< 0.001

Table 9: Augmented Dickey–Fuller Unit Root Tests

Series	Frequency	Test statistic	Lags	p-value
Prices	Daily	3.11	14	0.99
Prices	Weekly	3.38	8	0.99
Returns	Daily	-13.75	14	0.01
Returns	Weekly	-7.77	8	0.01

Table 10: Residual diagnostics for standardized residuals

	Ljung–Box (z_t)	Ljung–Box (z_t^2)	ARCH LM
Daily	0.510	0.638	0.648
Weekly	0.960	0.021	0.053

Table 11: Daily conditional volatility forecasts

Horizon	Forecasted σ
T+1	0.00873
T+2	0.00875
T+3	0.00876
T+4	0.00878
T+5	0.00879